

CD4066B CMOS 四路双向开关

1 特性

- 15V 数字或 $\pm 7.5V$ 峰峰值开关
- 工作电压为 15V 时
典型导通电阻为 125Ω
- 在 15V 信号输入范围内匹配的
开关导通电阻不超过 5Ω
- 在整个峰峰值信号范围内具有
平坦的导通电阻
- 高开关输出电压比：
 $f_{is} = 10kHz$ 、 $R_L = 1k\Omega$ 时典型值为 80dB
- 高度线性： $f_{is} = 1kHz$ 、 $V_{is} = 5V_{p-p}$ 、
 $V_{DD} - V_{SS} \geq 10V$ 、 $R_L = 10k\Omega$ 时失真典型值小于 0.5%
- 超低关断状态开关泄漏，从而产生较低的失调电流
和高有效关断状态电阻： $V_{DD} - V_{SS} = 10V$ 、 $T_A = 25^\circ C$ 时的典型值为 $10pA$
- 较高控制输入阻抗
(控制电路与信号电路相隔离)：
典型值为 $10^{12}\Omega$
- 低开关间串扰： $f_{is} = 8MHz$ 、 $R_L = 1k\Omega$ 时典型值为
- 50dB
- 匹配的控制输入到信号输出电容：可减少输出信号
瞬态
- 频率响应，
开启 = 40MHz (典型值)
- 针对 20V 下的静态电流进行了 100% 测试
- 5V、10V 和 15V 参数额定值

2 应用

- 模拟信号开关和多路复用：信号门控、调制器、静
噪控制、解调器、斩波器、换向开关
- 数字信号开关和多路复用
- 模数和数模转换
- 频率、阻抗、相位和模拟信号增益的数字控制
- 楼宇自动化

3 说明

CD4066B 器件是一款用于模拟或数字信号传输或多路复用的四路双向开关。该器件与 CD4016B 器件引脚对引脚兼容，但导通状态电阻低得多。此外，导通状态电阻在整个信号输入范围内相对恒定。

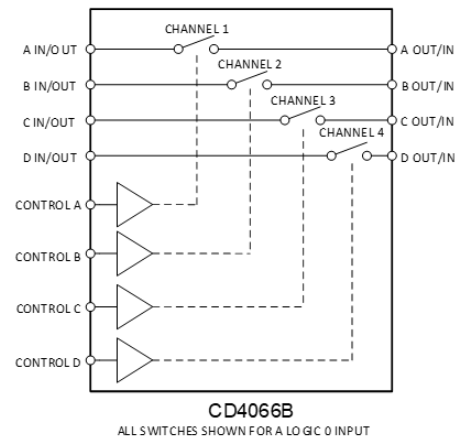
CD4066B 器件包含四个双向开关，每个开关都具有独立的控件。3V 至 18V 的宽工作电源电压范围支持用于广泛的应用。与单通道开关相比，优点包括峰值输入信号电压摆幅等于最大电源电压以及在输入信号范围内具有更恒定的导通状态阻抗。而对于采样保持应用，建议使用 CD4016B 器件。

封装信息

器件型号	封装 ⁽¹⁾	封装尺寸 ⁽²⁾
CD4066B	N (PDIP , 14)	19.3mm × 9.4mm
	D (SOIC , 14)	8.65mm × 6mm
	NS (SOP , 14)	10.2mm × 7.8mm
	PW (TSSOP , 14)	5mm × 6.4mm

(1) 有关更多信息，请参阅节 11

(2) 封装尺寸 (长 × 宽) 为标称值，并包括引脚 (如适用)。



通过数字控制逻辑进行双向信号传输



Table of Contents

1 特性	1	7.3 Feature Description.....	14
2 应用	1	7.4 Device Functional Modes.....	15
3 说明	1	8 Application and Implementation	16
4 Pin Configuration and Functions	3	8.1 Application Information.....	16
5 Specifications	4	8.2 Typical Application.....	16
5.1 Absolute Maximum Ratings.....	4	8.3 Power Supply Recommendations.....	17
5.2 ESD Ratings.....	4	8.4 Layout.....	17
5.3 Recommended Operating Conditions.....	4	9 Device and Documentation Support	19
5.4 Thermal Information.....	5	9.1 接收文档更新通知.....	19
5.5 Electrical Characteristics.....	5	9.2 支持资源.....	19
5.6 Switching Characteristics.....	8	9.3 Trademarks.....	19
5.7 Typical Characteristics.....	9	9.4 静电放电警告.....	19
6 Parameter Measurement Information	10	9.5 术语表.....	19
7 Detailed Description	14	10 Revision History	19
7.1 Overview.....	14	11 Mechanical, Packaging, and Orderable Information	19
7.2 Functional Block Diagram.....	14		

4 Pin Configuration and Functions

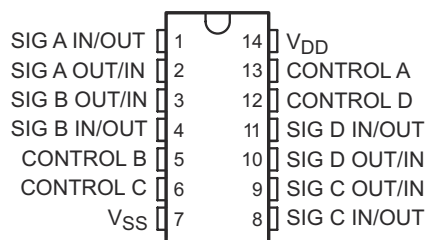


图 4-1. N, J, D, NS, or PW Packages 14-Pin PDIP, CDIP, SOIC, SOP, or TSSOP (Top View)

表 4-1. Pin Functions

PIN		TYPE ⁽¹⁾	DESCRIPTION
NAME	NO.		
SIG A IN/OUT	1	I/O	Input/Output for Switch A
SIG A OUT/IN	2	I/O	Output/Input for Switch A
SIG B OUT/IN	3	I/O	Output/Input for Switch B
SIG B IN/OUT	4	I/O	Input/Output for Switch B
CONTROL B	5	I	Control pin for Switch B
CONTROL C	6	I	Control pin for Switch C
V _{SS}	7	—	Low Voltage Power Pin
SIG C IN/OUT	8	I/O	Input/Output for Switch C
SIG C OUT/IN	9	I/O	Output/Input for Switch C
SIG D OUT/IN	10	I/O	Output/Input for Switch D
SIG D IN/OUT	11	I/O	Input/Output for Switch D
CONTROL D	12	I	Control Pin for D
CONTROL A	13	I	Control Pin for A
V _{DD}	14	—	Power Pin

(1) I = input, O = output

5 Specifications

5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾ ⁽²⁾

		MIN	MAX	UNIT
$V_{DD} - V_{SS}$	Supply voltage		20	V
V_{DD}		- 0.5	20	V
V_{SS}		- 20	0.5	V
I_{SEL} or I_{EN}	Logic control input pin current ($\overline{EN}$, Ax, SELx)	- 30	30	mA
V_S or V_D	Source or drain voltage (Sx, D)	$V_{SS} - 0.5$	$V_{DD} + 0.5$	V
I_S or I_D (CONT)	Source or drain continuous current (Sx, D)	- 20	20	mA
T_J	Junction temperature		150	°C
T_{stg}	Storage temperature	- 65	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Rating* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Condition*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) All voltages are with respect to ground, unless otherwise specified.

5.2 ESD Ratings

		VALUE	UNIT
$V_{(ESD)}$ Electrostatic discharge	Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins ⁽¹⁾	±500	V
	Charged device model (CDM), per JEDEC specification JESD22-C101, all pins ⁽²⁾	±1500	

(1) JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.

(2) JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.

5.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

		MIN	NOM	MAX	UNIT
$V_{DD} - V_{SS}$ ⁽¹⁾	Power supply voltage differential	3		18	V
V_{DD}	Positive power supply voltage	3		18	V
V_S or V_D	Signal path input/output voltage (source or drain pin) (Sx, D)	V_{SS}		V_{DD}	V
V_{SEL} or V_{EN}	Address or enable pin voltage	0		V_{DD}	V
I_S or I_D (CONT)	Source or drain continuous current (Sx, D)	- 10		10	mA
T_A	Ambient temperature	- 55		125	°C

(1) V_{DD} and V_{SS} can be any value as long as $3V \leq (V_{DD} - V_{SS}) \leq 24V$, and the minimum V_{DD} is met.

5.4 Thermal Information

THERMAL METRIC ⁽¹⁾		CD406x				UNIT
		N (PDIP)	D (SOIC)	NS (SO)	PW (TSSOP)	
		14 PINS	14 PINS	14 PINS	14 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	93.7	109.7	112.4	101.8	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	72.5	69.4	70.4	44.3	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	68.0	67.9	76.4	68.2	°C/W
Ψ_{JT}	Junction-to-top characterization parameter	50.3	25.8	28.9	3.2	°C/W
Ψ_{JB}	Junction-to-board characterization parameter	67.3	67.1	75.4	67.6	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

5.5 Electrical Characteristics

Over operating free-air temperature range, $V_{SUPPLY} = \pm 5V$, and $R_L = 100 \Omega$, (unless otherwise noted)⁽¹⁾

PARAMETER			TEST CONDITIONS		MIN	TYP	MAX	UNIT
SIGNAL INPUTS (V _{IS}) AND OUTPUTS (V _{OS})								
V _{OS}	Switch output voltage		V _{DD} = 5V V _{is} = 0V				0.4	V
			V _{DD} = 5V V _{is} = 5V				4.6	V
			V _{DD} = 10V V _{is} = 0V				0.5	V
			V _{DD} = 10V V _{is} = 10V				9.5	V
			V _{DD} = 15V V _{is} = 0V				1.5	V
			V _{DD} = 15V V _{is} = 15V				13.5	V
Δ R _{ON}	On-state resistance difference between any two switches		R _L = 10k Ω , V _C = V _{DD}	V _{DD} = 5V			15	Ω
	On-state resistance difference between any two switches	On-state resistance difference between any two switches		V _{DD} = 10V			10	
	On-state resistance difference between any two switches	On-state resistance difference between any two switches		V _{DD} = 15V			5	
V _{IHC}	Control input, high voltage		See Figure 7	V _{DD} = 5V			3.5	V
				V _{DD} = 10V			7	V
				V _{DD} = 15V			11	V

5.5 Electrical Characteristics (续)

Over operating free-air temperature range, $V_{\text{SUPPLY}} = \pm 5\text{V}$, and $R_L = 100\ \Omega$, (unless otherwise noted)⁽¹⁾

PARAMETER		TEST CONDITIONS				MIN	TYP	MAX	UNIT
	Maximum control input repetition rate	V _{IN} = V _{DD} , C _L = 50pF, R _L = 1kΩ V _C = 10V (square wave centered on 5V), t _r , t _f = 20ns, V _{os} = 1/2V _{os} at 1kHz	V _{DD} = 5V			6			MHz
			V _{DD} = 10V			9			
			V _{DD} = 15V			9.5			
C _{IN}	Input Capacitance					5	7.5		pF
I _{IS}	Switch input current	V _{DD} = 5V V _{is} = 0V	T _A = - 55°C			0.64			mA
			T _A = - 40°C			0.61			
			T _A = 25°C			0.51			
			T _A = 85°C			0.42			
			T _A = 125°C			0.36			
		V _{DD} = 5V V _{is} = 5V	T _A = - 55°C			-0.64			mA
			T _A = - 40°C			-0.61			
			T _A = 25°C			-0.51			
			T _A = 85°C			-0.42			
			T _A = 125°C			-0.36			
		V _{DD} = 10V V _{is} = 0V	T _A = - 55°C			1.6			mA
			T _A = - 40°C			1.5			
			T _A = 25°C			1.3			
			T _A = 85°C			1.1			
			T _A = 125°C			0.9			
		V _{DD} = 10V V _{is} = 10V	T _A = - 55°C			-1.6			mA
			T _A = - 40°C			-1.5			
			T _A = 25°C			-1.3			
			T _A = 85°C			-1.1			
			T _A = 125°C			-0.9			
		V _{DD} = 15V V _{is} = 0V	T _A = - 55°C			4.2			mA
			T _A = - 40°C			4			
			T _A = 25°C			3.4			
			T _A = 85°C			2.8			
			T _A = 125°C			2.4			
		V _{DD} = 15V V _{is} = 15V	T _A = - 55°C			-4.2			mA
			T _A = - 40°C			-4			
			T _A = 25°C			-3.4			
			T _A = 85°C			-2.8			
			T _A = 125°C			-2.4			

5.5 Electrical Characteristics (续)

Over operating free-air temperature range, $V_{\text{SUPPLY}} = \pm 5\text{V}$, and $R_L = 100\ \Omega$, (unless otherwise noted)⁽¹⁾

PARAMETER		TEST CONDITIONS				MIN	TYP	MAX	UNIT
I_{DD}	Quiescent Device Current All switches OFF	$V_{\text{IS}} = 0 \text{ to } 5\text{V}$ $V_{\text{DD}} = 5\text{V}$	$T_A = -55^\circ\text{C}$					5	μA
			$T_A = -40^\circ\text{C}$					5	
			$T_A = 25^\circ\text{C}$				4	6	
			$T_A = 85^\circ\text{C}$					7	
			$T_A = 125^\circ\text{C}$					7.5	
		$V_{\text{IS}} = 0 \text{ to } 10\text{V}$ $V_{\text{DD}} = 10\text{V}$	$T_A = -55^\circ\text{C}$					6	
			$T_A = -40^\circ\text{C}$					6	
			$T_A = 25^\circ\text{C}$				5	7	
			$T_A = 85^\circ\text{C}$					8	
			$T_A = 125^\circ\text{C}$					9	
		$V_{\text{IS}} = 0 \text{ to } 15\text{V}$ $V_{\text{DD}} = 15\text{V}$	$T_A = -55^\circ\text{C}$					7	
			$T_A = -40^\circ\text{C}$					7.5	
			$T_A = 25^\circ\text{C}$				5.5	8	
			$T_A = 85^\circ\text{C}$					9	
			$T_A = 125^\circ\text{C}$					10	
		$V_{\text{IS}} = 0 \text{ to } 20\text{V}$ $V_{\text{DD}} = 20\text{V}$	$T_A = -55^\circ\text{C}$					8.5	
			$T_A = -40^\circ\text{C}$					8.5	
			$T_A = 25^\circ\text{C}$				6.5	9	
			$T_A = 85^\circ\text{C}$					10	
			$T_A = 125^\circ\text{C}$					11	
r_{ON}	ON Resistance r_{ON} Max	to $(V_{\text{DD}} + V_{\text{SS}})/2$, V_{C} $= V_{\text{DD}}$, $R_L = 10\text{k}\Omega$ returned $V_{\text{IS}} = V_{\text{SS}}$ to V_{DD}	$V_{\text{DD}} = 5\text{V}$	$T_A = -55^\circ\text{C}$				800	Ω
				$T_A = -40^\circ\text{C}$				850	
				$T_A = 25^\circ\text{C}$			470	1050	
				$T_A = 85^\circ\text{C}$				1200	
				$T_A = 125^\circ\text{C}$				1300	
			$V_{\text{DD}} = 10\text{V}$	$T_A = -55^\circ\text{C}$				310	
				$T_A = -40^\circ\text{C}$				330	
				$T_A = 25^\circ\text{C}$			180	400	
				$T_A = 85^\circ\text{C}$				500	
				$T_A = 125^\circ\text{C}$				500	
			$V_{\text{DD}} = 15\text{V}$	$T_A = -55^\circ\text{C}$				200	
				$T_A = -40^\circ\text{C}$				210	
				$T_A = 25^\circ\text{C}$			125	240	
				$T_A = 85^\circ\text{C}$				300	
				$T_A = 125^\circ\text{C}$				320	

5.5 Electrical Characteristics (续)

Over operating free-air temperature range, $V_{\text{SUPPLY}} = \pm 5\text{V}$, and $R_L = 100\ \Omega$, (unless otherwise noted)⁽¹⁾

PARAMETER			TEST CONDITIONS				MIN	TYP	MAX	UNIT
V _{ILC}	Control input, low voltage (max)		I _{is} < 10 μA, V _{is} = V _{SS} , V _{OS} = V _{DD} , and V _{is} = V _{DD} , V _{OS} = V _{SS}	V _{DD} = 5V	T _A = -55°C			1	V	
					T _A = -40°C			1		
					T _A = 25°C			1		
					T _A = 85°C			1		
					T _A = 125°C			1		
				V _{DD} = 10V	T _A = -55°C			1		
					T _A = -40°C			1		
					T _A = 25°C			1		
					T _A = 85°C			1		
					T _A = 125°C			1		
				V _{DD} = 15V	T _A = -55°C			1		
					T _A = -40°C			1		
					T _A = 25°C			1		
					T _A = 85°C			1		
					T _A = 125°C			1		
I _{IN}	Input current (max)		V _{is} ≤ V _{DD} , V _{DD} - V _{SS} = 18V, V _{CC} ≤ V _{DD} - V _{SS} , V _{DD} = 18V	T _A = -55°C			-0.8	0.8	μA	
				T _A = -40°C			-0.8	0.8		
				T _A = 25°C			-0.7	±0.2		0.7
				T _A = 85°C			-0.6	0.6		
				T _A = 125°C			-0.55	0.55		

(1) Peak-to-Peak voltage symmetrical about $(V_{\text{DD}} - V_{\text{EE}}) / 2$.

5.6 Switching Characteristics

$T_A = 25^\circ\text{C}$

PARAMETER	FROM	TO	TEST CONDITIONS	V_{CC}	MIN	TYP	MAX	UNIT
t_{pd}	Signal input	Signal output	$V_{\text{IN}} = V_{\text{DD}}, t_r, t_f = 20\text{ns}, C_L = 50\text{pF}, R_L = 1\text{k}\ \Omega$	5V		20	40	ns
				10V		10	20	
				15V		7	15	
t_{plh}	Signal input	Signal output	$V_{\text{IN}} = V_{\text{DD}}, t_r, t_f = 20\text{ns}, C_L = 50\text{pF}, R_L = 1\text{k}\ \Omega$	5V		35	70	ns
				10V		20	40	
				15V		15	30	
t_{phl}	Signal input	Signal output	$V_{\text{IN}} = V_{\text{DD}}, t_r, t_f = 20\text{ns}, C_L = 50\text{pF}, R_L = 1\text{k}\ \Omega$	5V		35	70	ns
				10V		20	40	
				15V		15	30	

5.7 Typical Characteristics

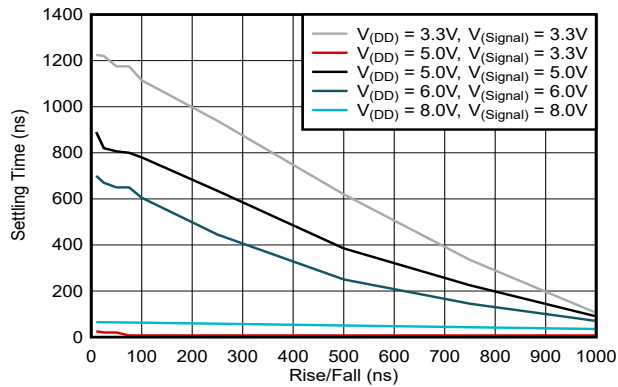


图 5-1. System Settling Time vs Signal Rise/Fall Time

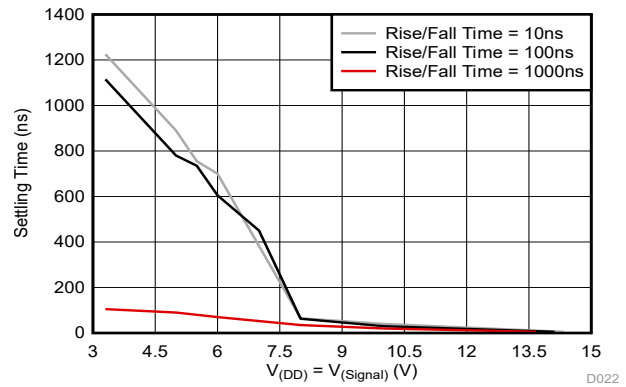


图 5-2. System Settling Time vs Signal Voltage

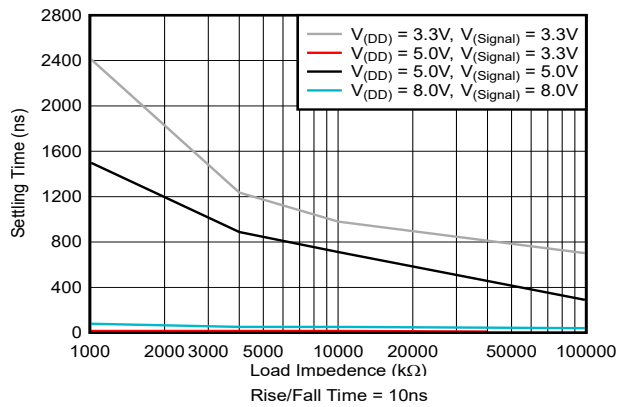


图 5-3. System Settling Time vs Signal Voltage

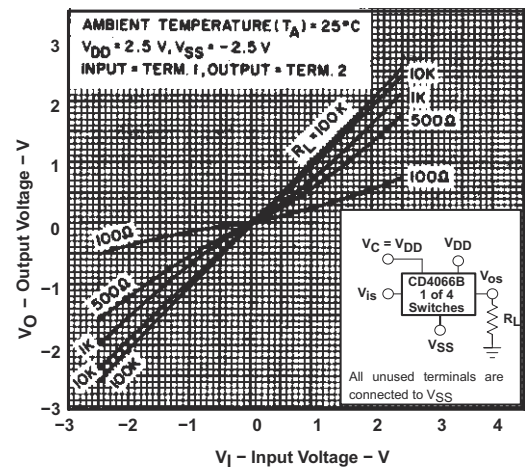


图 5-4. Typical ON Characteristics for 1 of 4 Channels

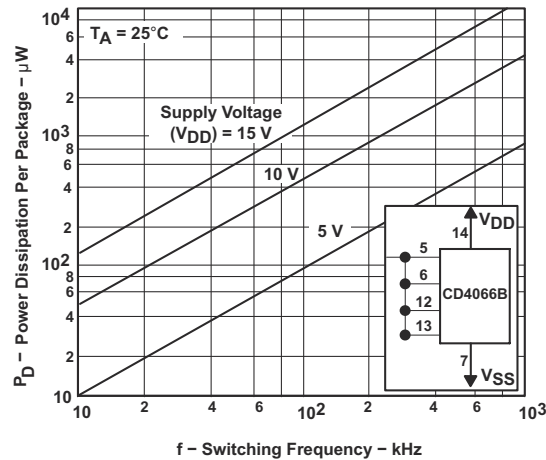
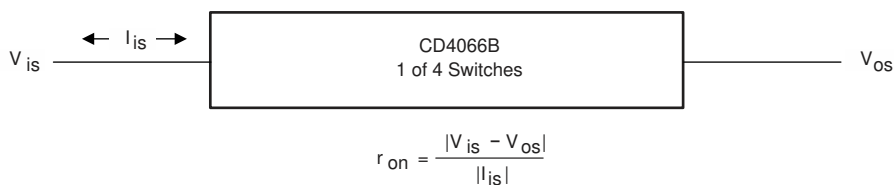


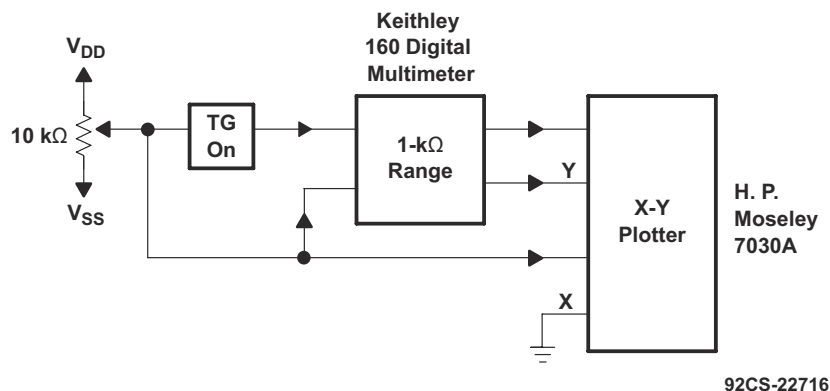
图 5-5. Power Dissipation per Package vs Switching Frequency

6 Parameter Measurement Information



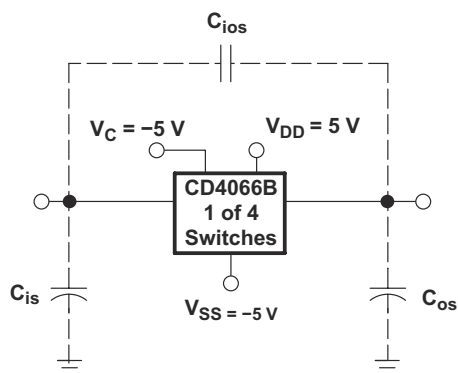
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图 6-1. Determination of r_{on} as a Test Condition for Control-Input High-Voltage (V_{IHC}) Specification



92CS-22716

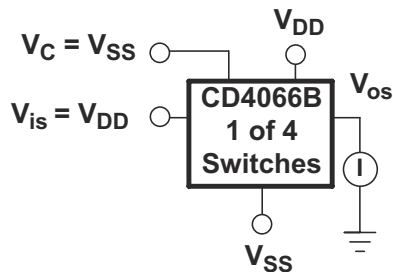
图 6-2. Channel On-State Resistance Measurement Circuit



92CS-30921

Measured on Boonton capacitance bridge, model 75a (1 MHz);
test-fixture capacitance nulled out.

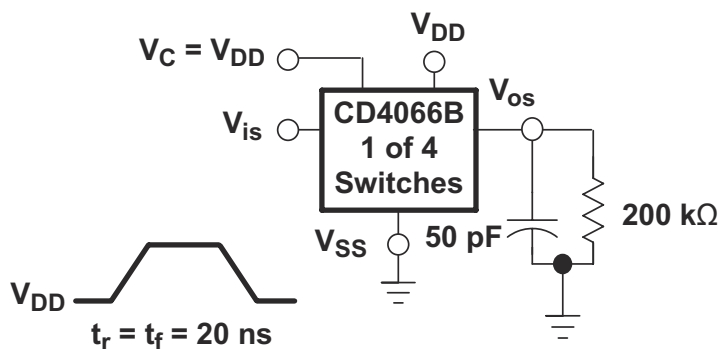
图 6-3. Typical On Characteristics for One of Four Channels



92CS-30922

All unused terminals are connected to V_{SS} .

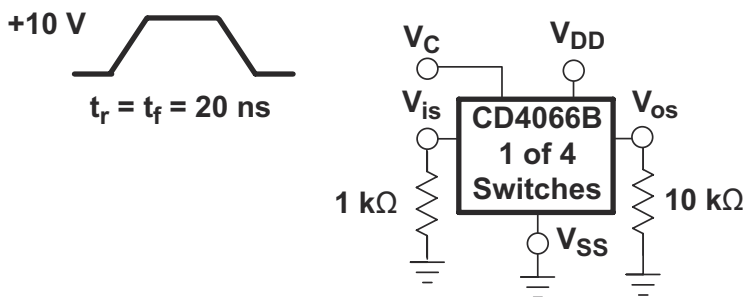
图 6-4. Off-Switch Input or Output Leakage



92CS-30923

All unused terminals are connected to V_{SS} .

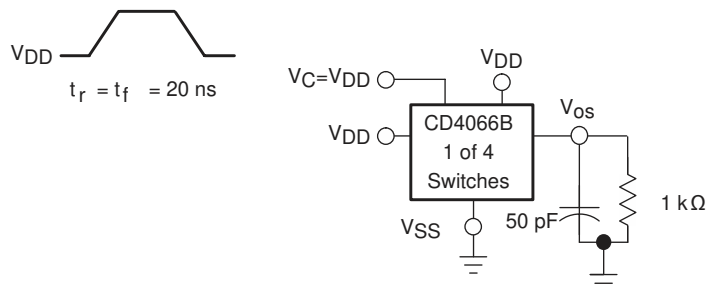
图 6-5. Propagation Delay Time Signal Input (V_{is}) to Signal Output (V_{os})



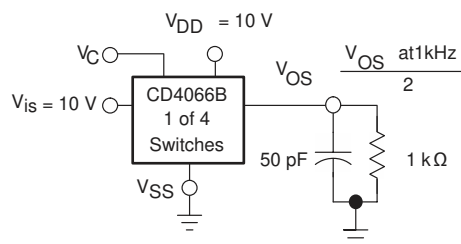
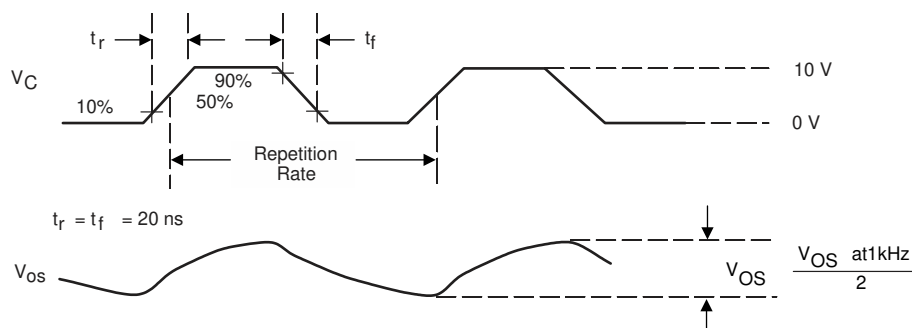
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All unused terminals are connected to V_{SS} .

图 6-6. Crosstalk-Control Input to Signal Output

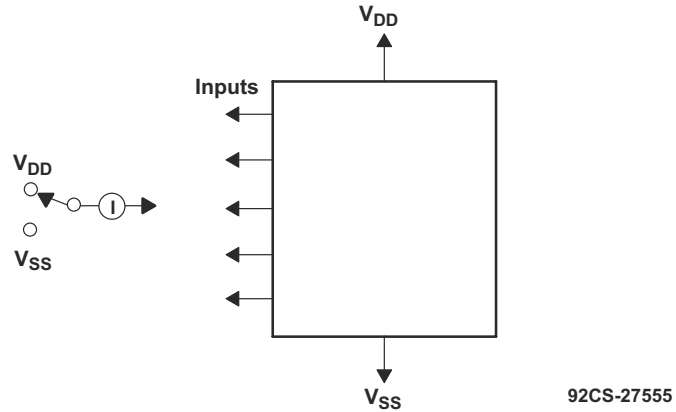


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All unused pins are connected to V_{SS} .Delay is measured at V_{OS} level of +10% from ground (turn-on) or on-state output level (turn-off).**图 6-7. Propagation Delay, t_{PLH} , t_{PHL} Control-Signal Output**

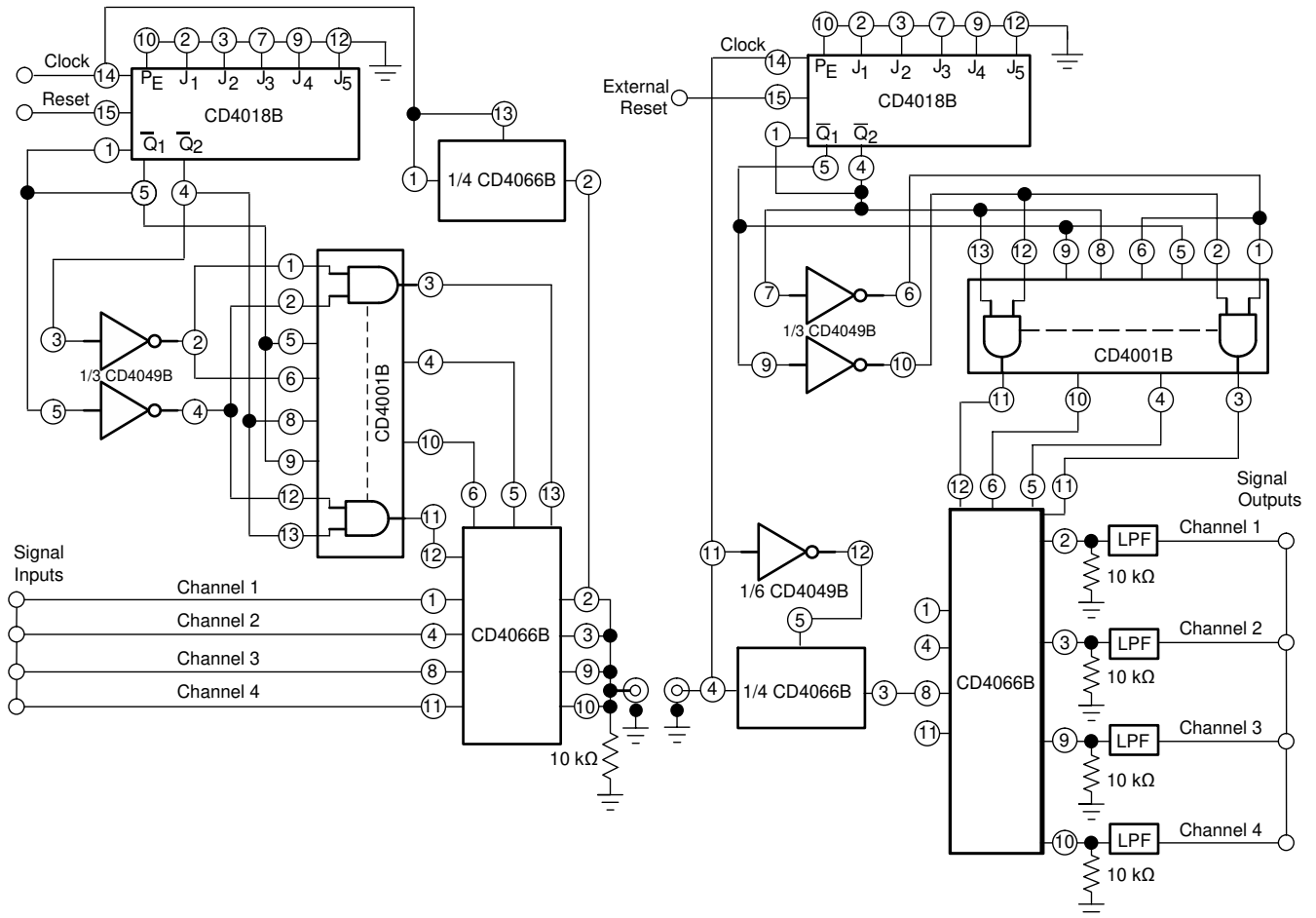
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All unused pins are connected to V_{SS} .**图 6-8. Maximum Allowable Control-Input Repetition Rate**



Measure inputs sequentially to both V_{DD} and V_{SS} . Connect all unused inputs to either V_{DD} or V_{SS} . Measure control inputs only.

图 6-9. Input Leakage-Current Test Circuit



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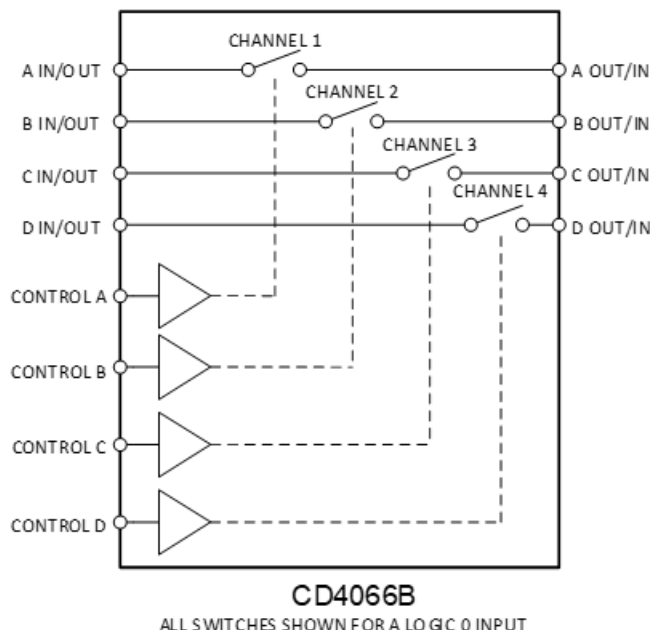
图 6-10. Four-Channel PAM Multiplex System Diagram

7 Detailed Description

7.1 Overview

CD4066B has four independent digitally controlled analog switches with a bias voltage of V_{SS} to allow for different voltage levels to be used for low output. Both the p and n devices in a given switch are biased on or off simultaneously by the control signal. As shown in 图 7-1, the well of the n-channel device on each switch is tied to either the input (when the switch is on) or to V_{SS} (when the switch is off). Thus, when the control of the device is low, the output of the switch goes to V_{SS} and when the control is high the output of the device goes to V_{DD} .

7.2 Functional Block Diagram



- A. All control inputs are protected by the CMOS protection network.
- B. All p substrates are connected to V_{DD} .
- C. Normal operation control-line biasing: switch on (logic 1), $V_C = V_{DD}$; switch off (logic 0), $V_C = V_{SS}$.
- D. Signal-level range: $V_{SS} \leq V_{is} \leq V_{DD}$.

图 7-1. Schematic Diagram of One-of-Four Identical Switches and Associated Control Circuitry

7.3 Feature Description

Each switch has different control pins, which allows for more options for the outputs. Bias Voltage allows the device to output a voltage other than 0V when the device control is low. The CD4066B has a large absolute maximum voltage for V_{DD} of 20V.

7.4 Device Functional Modes

表 7-1 lists the functions of this device.

表 7-1. Function Table

INPUTS		OUTPUT
SIG IN/OUT	CONTROL	SIG OUT/IN
H	H	H
L	H	L
X	L	Hi-Z

8 Application and Implementation

备注

以下应用部分中的信息不属于 TI 器件规格的范围，TI 不担保其准确性和完整性。TI 的客户应负责确定器件是否适用于其应用。客户应验证并测试其设计，以确保系统功能。

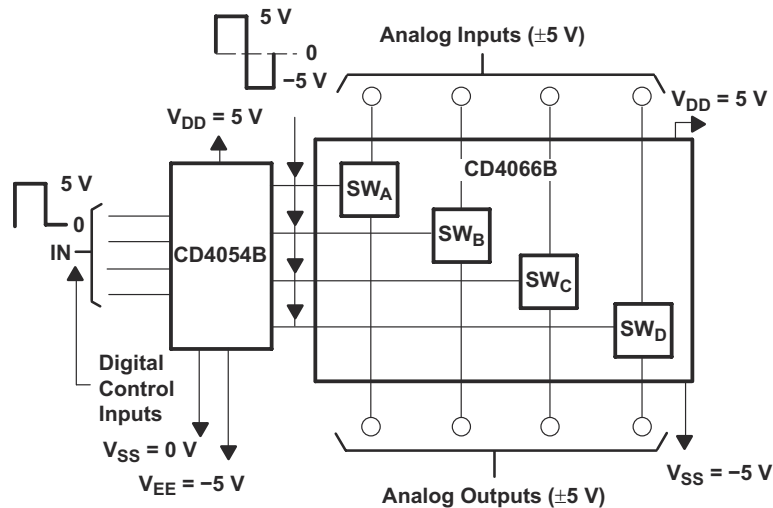
8.1 Application Information

In applications that employ separate power sources to drive V_{DD} and the signal inputs, the V_{DD} current capability should exceed V_{DD}/R_L (R_L = effective external load of the four CD4066B device bilateral switches). This provision avoids any permanent current flow or clamp action on the V_{DD} supply when power is applied or removed from the CD4066B device.

In certain applications, the external load-resistor current can include both V_{DD} and signal-line components. To avoid drawing V_{DD} current when switch current flows into pins 1, 4, 8, or 11, the voltage drop across the bidirectional switch must not exceed 0.8V (calculated from r_{on} values shown).

No V_{DD} current flows through R_L if the switch current flows into pins 2, 3, 9, or 10.

8.2 Typical Application



92CS-30927

图 8-1. Bidirectional Signal Transmission Through Digital Control Logic

8.2.1 Design Requirements

This device uses CMOS technology and has balanced output drive. Avoid bus contention because it can drive currents in excess of maximum limits. The high drive also creates fast edges into light loads, so consider routing and load conditions to prevent ringing.

8.2.2 Detailed Design Procedure

- Recommended input conditions:
 - For rise time and fall time specifications, see $\Delta t / \Delta v$ in [Recommended Operating Conditions](#).
 - For specified high and low levels, see V_{IH} and V_{IL} in [Recommended Operating Conditions](#).
- Recommended output conditions:
 - Load currents should not exceed $\pm 10\text{mA}$.

8.2.3 Application Curve

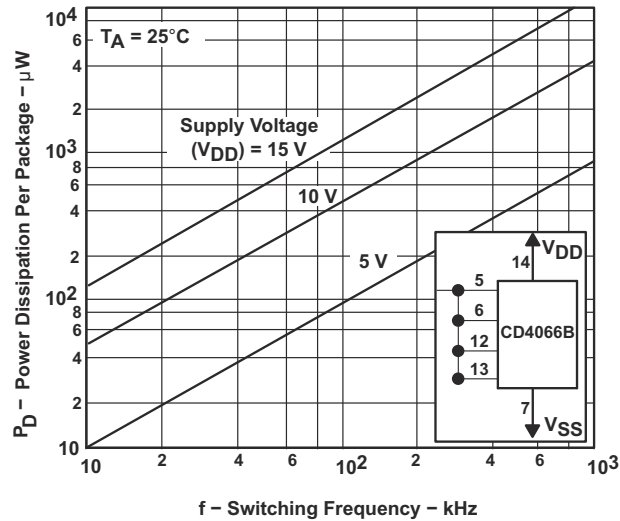


图 8-2. Power Dissipation vs. Switching Frequency

8.3 Power Supply Recommendations

The power supply can be any voltage between the MIN and MAX supply voltage rating located in [Recommended Operating Conditions](#).

Each VCC pin should have a good bypass capacitor to prevent power disturbance. For devices with a single supply, $0.1\mu\text{F}$ is recommended; if there are multiple VCC pins, then $0.01\mu\text{F}$ or $0.022\mu\text{F}$ is recommended for each power pin. It is acceptable to parallel multiple bypass capacitors to reject different frequencies of noise. A $0.1\mu\text{F}$ and a $1\mu\text{F}$ are commonly used in parallel. The bypass capacitor should be installed as close to the power pin as possible for best results.

8.4 Layout

8.4.1 Layout Guidelines

When using multiple bit logic devices inputs must never float.

In many cases, functions or parts of functions of digital logic devices are unused, for example, when only two inputs of a triple-input *and* gate are used or only 3 of the 4 buffer gates are used. Such input pins must not be left unconnected because the undefined voltages at the outside connections result in undefined operational states. All unused inputs of digital logic devices must be connected to a high or low bias to prevent them from floating. The logic level that should be applied to any particular unused input depends on the function of the device. Generally they are tied to GND or VCC, whichever makes more sense or is more convenient. It is generally acceptable to float outputs, unless the part is a transceiver. If the transceiver has an output enable pin, it disables the output section of the part when asserted. This does not disable the input section of the I/Os, so they cannot float when disabled.

8.4.2 Layout Example

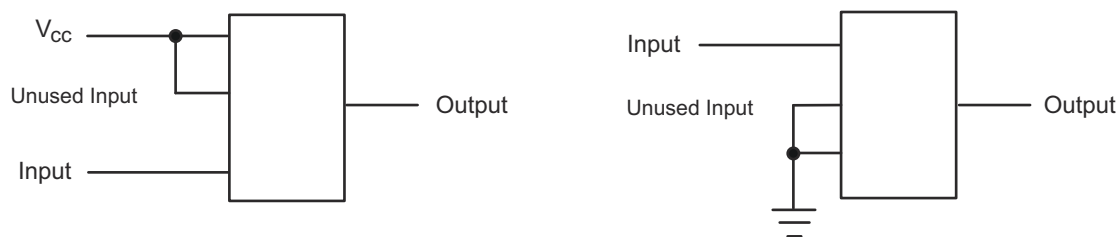


图 8-3. Diagram for Unused Inputs

9 Device and Documentation Support

9.1 接收文档更新通知

要接收文档更新通知，请导航至 ti.com 上的器件产品文件夹。点击 [通知](#) 进行注册，即可每周接收产品信息更改摘要。有关更改的详细信息，请查看任何已修订文档中包含的修订历史记录。

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[TI E2E™ 中文支持论坛](#) 是工程师的重要参考资料，可直接从专家处获得快速、经过验证的解答和设计帮助。搜索现有解答或提出自己的问题，获得所需的快速设计帮助。

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ESD 的损坏小至导致微小的性能降级，大至整个器件故障。精密的集成电路可能更容易受到损坏，这是因为非常细微的参数更改都可能会导致器件与其发布的规格不相符。

9.5 术语表

[TI 术语表](#) 本术语表列出并解释了术语、首字母缩略词和定义。

10 Revision History

注：以前版本的页码可能与当前版本的页码不同

Changes from Revision I (May 2024) to Revision J (August 2024)	Page
• Added Settling Time plots.....	9

Changes from Revision H (January 2020) to Revision I (May 2024)	Page
• 更改了 封装信息 表以包含封装引线.....	1
• 通篇更改了表格、图和交叉参考的编号格式.....	1
• 删除了数据表中的 J (CDIP , 14) 封装.....	1
• Changed max and typ IDD for lower supply voltages.....	5
• Changed VIL from 2V to 1V across supply.....	5

Changes from Revision G (June 2017) to Revision H (January 2020)	Page
• Added Junction Temperature details to the <i>Absolute Maximum Ratings</i> table.....	4

11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
CD4066BE	ACTIVE	PDIP	N	14	25	RoHS & Green	NIPDAU	N / A for Pkg Type	-55 to 125	CD4066BE	Samples
CD4066BEE4	ACTIVE	PDIP	N	14	25	RoHS & Green	NIPDAU	N / A for Pkg Type	-55 to 125	CD4066BE	Samples
CD4066BF	ACTIVE	CDIP	J	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	CD4066BF	Samples
CD4066BF3A	ACTIVE	CDIP	J	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	CD4066BF3A	Samples
CD4066BM96	ACTIVE	SOIC	D	14	2500	RoHS & Green	NIPDAU SN	Level-1-260C-UNLIM	-55 to 125	CD4066BM	Samples
CD4066BNSR	ACTIVE	SO	NS	14	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4066B	Samples
CD4066BPWR	ACTIVE	TSSOP	PW	14	2000	RoHS & Green	NIPDAU SN	Level-1-260C-UNLIM	-55 to 125	CM066B	Samples
JM38510/05852BCA	ACTIVE	CDIP	J	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	JM38510/ 05852BCA	Samples
M38510/05852BCA	ACTIVE	CDIP	J	14	25	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	JM38510/ 05852BCA	Samples

(1) The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBsolete: TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".

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Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

⁽⁵⁾ Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

⁽⁶⁾ Lead finish/Ball material - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

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OTHER QUALIFIED VERSIONS OF CD4066B, CD4066B-MIL :

- Catalog : [CD4066B](#)
- Automotive : [CD4066B-Q1](#), [CD4066B-Q1](#)
- Military : [CD4066B-MIL](#)

NOTE: Qualified Version Definitions:

- Catalog - TI's standard catalog product
- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects
- Military - QML certified for Military and Defense Applications

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